

Report: Learning Path Creator

Introduction: The Learning Path Creator is a web-based platform designed to facilitate a community-driven approach to learning. The application empowers users to create, share, and explore curated lists of educational resources, referred to as "learning paths." The primary purpose is to provide a collaborative space where learners can benefit from the collective knowledge of the community. Users can compile resources, vote on the quality of learning paths, and engage in collaborative learning experiences.

Key functionalities include resource compilation, community-driven curation through voting, and the ability for registered users to clone and personalize existing learning paths. The application targets a diverse audience, including individual learners seeking structured paths on specific topics and educators/experts who wish to share their expertise through comprehensive learning paths.

With features such as user registration, authentication, profile management, and an innovative voting system, the Learning Path Creator aims to enhance the accessibility and customization of learning experiences, making education more tailored to individual needs.

Project Features:

1. User Registration and Authentication:

- Users can register for a new account, providing necessary details.
- Authentication is handled securely using PHP sessions and password hashing.

2. Learning Path Creation:

- Users can create custom learning paths by adding various learning modules, such as articles, videos, or quizzes.
- Each module contains a title, description, and a link to the educational resource.

3. Path Collaboration:

- Users can collaborate on learning paths, enabling group learning experiences.
- Collaborators can suggest modifications or additions to the learning path.

4. Interactive User Interface:

- The user interface is built using HTML and JavaScript, providing a responsive and intuitive experience.
- Drag-and-drop functionality is implemented to allow easy rearrangement of learning modules within a path.

5. Path Progress Tracking:

- Users can track their progress through a learning path, marking completed modules.
- Progress is visually represented to enhance the user experience.

6. Search and Discovery:

- A search functionality helps users discover existing learning paths created by others.
- Tags and categories assist in organizing and filtering learning path.

Frontend (Presentation Layer):

- **HTML, CSS, and JavaScript:** The core building blocks for creating the user interface, structuring content, and providing interactivity.

Backend (Application Logic):

- **PHP:** The server-side scripting language responsible for processing user requests, managing authentication, and handling business logic.

User Authentication: User authentication is a crucial aspect of the Learning Path Creator application, ensuring that only registered and authorized users can access certain functionalities. The process involves verifying the identity of users based on their provided credentials. Here's how user authentication is handled:

1. User Registration:

- Users can register for an account by providing a unique username, a valid email address, and a secure password.
- The application securely stores user passwords using hashing techniques to enhance security.

2. Login:

- Registered users can log in by entering their username/email and password.
- The entered credentials are validated against the stored credentials in the database.

3. Session Management:

- Upon successful login, a session is initiated for the user, maintaining their authentication status throughout their interaction with the application.
- PHP's session handling mechanisms are employed to manage user sessions securely.

Authorization: Authorization controls access to specific functionalities or resources within the application. Different user roles may have varying levels of access. Here's how authorization is handled:

1. User Roles:

- The application may have different user roles, such as "user" and "admin," each with distinct permissions.
- User roles are stored in the database and assigned to users during the registration process.

2. Role-Based Access Control (RBAC):

- Authorization checks are performed based on the user's role to determine whether they have permission to perform specific actions.
- For example, only administrators may have the authority to delete learning paths.

3. Middleware:

- Middleware functions are implemented in the backend to intercept requests and ensure that the user making the request has the necessary permissions.
- Unauthorized users are redirected or presented with appropriate error messages.

Database Interaction: The Learning Path Creator application interacts with its MySQL database using PHP for server-side scripting. Employing My SQLi or PDO extensions ensures secure and parameterized SQL queries for actions like retrieval, insertion, and updates. The database schema, represented in a structured blueprint, defines tables, relationships, and data fields. This schema forms the foundation for organizing user data, learning paths, and other entities. Through this method, the application ensures efficient and secure communication with the database, promoting reliability and data integrity.

Conclusion: The Learning Path Creator project has successfully realized its vision of fostering collaborative learning through community-driven educational paths. The user-friendly interface, robust authentication mechanisms, and innovative features contribute to a seamless and engaging experience. The scalable architecture, code quality, and comprehensive documentation set a solid foundation for future enhancements. By empowering users to actively participate in creating and sharing educational content, the application not only meets its goals but also promises a positive impact on the way individuals approach learning. In conclusion, the Learning Path Creator stands as a testament to effective collaboration, innovation, and the potential for positive change in the realm of education.

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Video URL	https://vimeo.com/889743656?share=copy
GB Learn URL	https://f3450921.gblearn.com/comp1230/assignments/project/Learningpathcreator/
Tasks completed by member1	Front-end development, including HTML, CSS, and JavaScript.
	User interface design and layout.
	Implementation of responsive design for various screen sizes.
	Integration of third-party libraries or frameworks.
	Collaborating on the creation and enhancement of learning path views.

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Video URL	https://youtube.com/shorts/99KwqkEDs6U?si=Yuoz2T7JIJB60G_U
GB Learn URL	https://f3414811.gblearn.com/comp1230/assignments/project/Learningpathcreator/
Tasks completed by member2	Back-end development using PHP.
	Server-side scripting and database interactions.
	User authentication and authorization implementation
	Database schema design and maintenance.
	Error handling and validation mechanisms in PHP.

Full Name:	
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Video URL	
GB Learn URL	
Tasks completed by member3	Database queries optimization.
	Implementation of the collaborative path enhancement feature.
	Documentation creation, including the README file
	Code quality assurance through code reviews.
	Testing and bug fixing throughout the development cycle.